

SHI: Inspecting and Correcting Structured Hypotheses in Interactive Video Retrieval

Phuc-Khang Nguyen¹, Nguyen-Huy Le-Huu¹, Phuc Quang Vo¹, Danh-Phuong Nguyen¹, and Huy Dinh Nhat Truong¹

University of Science, Vietnam National University Ho Chi Minh City,
Ho Chi Minh City, Vietnam
{24120068, 24120060, 24120414, 24120504, 24120064}@student.hcmus.edu.vn

Abstract. Multi-event video retrieval under live competition constraints requires interpreting a textual narrative as an ordered sequence of retrievable events and aligning those events with occurrences within candidate videos. If users rely solely on query reformulation and result inspection, they cannot directly inspect how the system decomposes the narrative or selects an occurrence for each event. We present SHI, a human-in-the-loop video retrieval system that integrates two mechanisms for inspecting and correcting these decisions: (1) *Query Hypothesis*, which lets users inspect and revise an ordered event decomposition grounded in the original description; and (2) *EventTrail*, which exposes the temporal alignment of the complete event sequence within a candidate video. EventTrail presents alternative occurrences in the context of complete ordered paths, allowing users to retain the current match, select an alternative, or reject a temporal region. User selections are treated as temporal feedback: retaining or selecting an occurrence creates an anchor constraint, while rejecting a temporal region creates an exclusion constraint. SHI then re-decodes the complete alignment under these constraints.

Keywords: Interactive video retrieval · Video Browser Showdown · Temporal alignment · Human-in-the-loop · Multi-event search

1 Introduction

The Video Browser Showdown (VBS) evaluates interactive video retrieval systems through live search tasks over large video collections [18]. We present **SHI**, a system for VBS 2027 that combines multimodal retrieval, ranked-result browsing, and temporal alignment. SHI provides dedicated search workflows for Known-Item Search (KIS) and Ad-hoc Video Search (AVS), together with task-aware submission to the competition server. Its retrieval pipeline searches visual and textual evidence from frames, captions, optical character recognition (OCR), and speech transcripts, then fuses the retrieved candidates for interactive inspection. This paper focuses on the additional workflow SHI provides for KIS descriptions that contain several temporally ordered events.

Retrieval quality alone does not determine performance in an interactive search session. A post-evaluation of VBS 2022 found that the system with the strongest initial text-to-image retrieval did not obtain the highest overall score; browsing and image-based queries also contributed to task performance [15]. An analysis of interaction logs

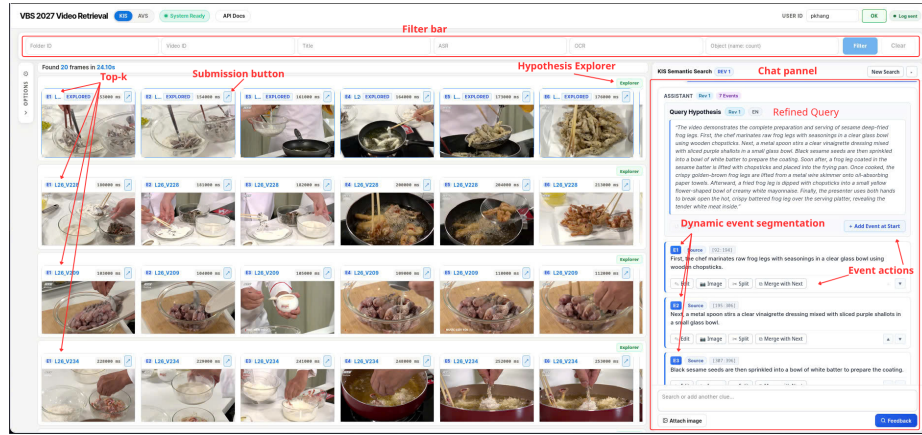


Fig. 1. SHI search interface. The main area presents ranked candidate videos and event-level matches. The right panel shows the current *Query Hypothesis* and provides controls for editing events, changing their structure, and adding image examples.

from two systems at VBS 2026 likewise found that operators repeatedly reformulated text queries and inspected the returned results [8]. SHI retains this ranked-grid workflow while giving the operator direct control over two system decisions that are consequential for multi-event KIS: how the description is divided into events and which occurrence is assigned to each event in a candidate video.

For a multi-event KIS description, SHI creates an ordered *Query Hypothesis* before search. The operator can edit, split, merge, or reorder its events and can attach an image example to an individual event. SHI searches each event against its multimodal indexes and constructs an event–frame score matrix for each candidate video. A temporal decoder uses these scores to find chronologically valid paths through the video, and the interface displays the resulting event matches alongside the ranked candidates.

When an event has been matched to the wrong occurrence, the operator can open *EventTrail* for that candidate. *EventTrail* presents the current path and complete-path alternatives rather than isolated frames. The operator may confirm the current occurrence, select an alternative, or exclude an incorrect temporal region. SHI then re-decodes the path from the retained event scores under the updated constraints. These local corrections do not require another corpus retrieval call.

The system contributions described in this paper are:

- an end-to-end VBS system that combines multimodal retrieval, ranked browsing, temporal alignment, and task-aware result submission;
- a structured multi-event KIS workflow in which the operator can edit the event sequence used for retrieval and inspect the corresponding matches in each candidate video; and
- *EventTrail*, which supports event-level confirmation, replacement, and rejection followed by constrained temporal re-decoding.

2 Related VBS Systems

VBS systems typically combine several retrieval modes with an interface for rapid result inspection. VISIONE provides textual, visual, and temporal search over a shared video index [1–3]. Vibro and diveXplore emphasize efficient browsing and visual exploration of large result sets [7, 16], while PraK builds its search interface on reusable video data services [12]. Recent versions of vitivr likewise combine multimodal query formulation with a redesigned retrieval engine [5, 4]. SHI follows this system pattern: multimodal retrieval produces a ranked grid from which the operator can inspect candidates, open temporal context, and submit an answer.

Several VBS systems support interaction beyond an initial query. Text-query re-formulation can use suggested concepts derived from early results [11]. Exquisitor combines conversational query revision with explicit relevance feedback on retrieved items [10, 17], and vitivr has investigated interfaces that make retrieval decisions easier to inspect [6]. The *Query Hypothesis* in SHI applies this interactive principle to multi-event KIS: the event sequence produced from the description remains visible and editable before a search is executed.

Temporal queries have been expressed through structured conditions [19], contextualized multimodal stages [5], and sequence chains [9]. Interactive moment retrieval has also used feedback to guide navigation among candidate moments [13]. SHI uses an ordered-event decoder to obtain a complete path for each candidate video. *EventTrail* makes that path an interaction object: the operator can inspect alternative occurrences and apply event-level constraints before the path is decoded again.

3 System Overview

SHI separates corpus preparation from online search. Offline jobs produce canonical frame metadata, specialist evidence, and retrieval indexes. The competition runtime opens these artifacts in read-only mode and provides separate workflows for multi-event KIS and AVS. Figure 2 shows the shared data layer and the two online paths.

3.1 Offline Evidence and Indexes

The VBS preparation profile currently targets the V3C collection [14]. Each sampled frame is stored with its `video_id`, internal `frame_id`, frame coordinate `frame_idx`, and `timestamp_ms`. SHI generates captions with Qwen3-VL-2B, extracts text with Florence-2, and detects objects with YOLOE. Caption, OCR, and object outputs remain separate artifacts. A deterministic *FrameContext* joins their usable text for context retrieval without discarding the specialist evidence. Speech is stored as timestamped ASR segments rather than being treated as frame-native evidence.

SHI builds three dense indexes: a SigLIP2 index over visual frames, a BGE-M3 index over *FrameContext*, and a BGE-M3 index over ASR segments. When frame-level scores or candidates are required, SHI projects ASR evidence through canonical video and timestamp mappings. A separate BM25 index covers title, caption, OCR, and ASR fields. Index creation and enrichment are offline operations; the online service does not regenerate an artifact when a user submits a query.

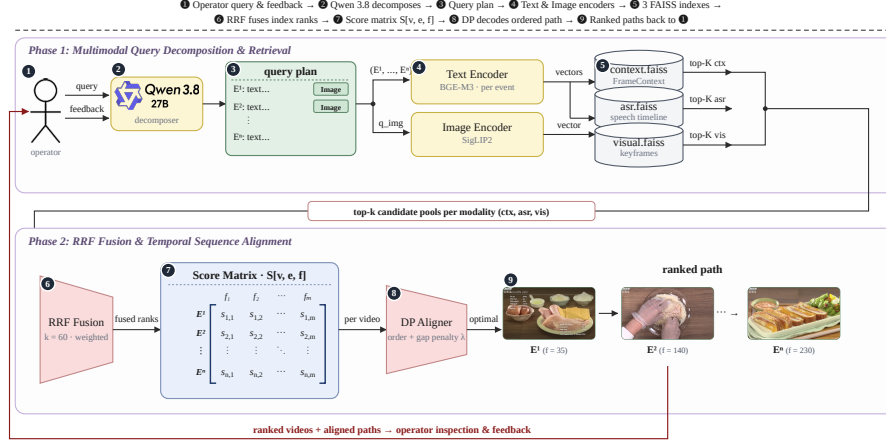


Fig. 2. Retrieval pipeline. The operator provides a query and interactive feedback to Qwen 3.8, which decomposes the request into an ordered query plan $(E^1, \dots, E^n)$. Each event is embedded by a text encoder (BGE-M3) and, optionally, an image encoder (SigLIP2) if a reference image is attached. The resulting vectors are matched against three FAISS indexes (frame context, ASR/speech timeline, and visual keyframes); the per-index top- k results are fused by Reciprocal Rank Fusion ($k = 60$) into a score matrix $S[v, e, f]$. A dynamic-programming aligner then decodes, per candidate video v , the best-ordered path through the events subject to a temporal gap penalty λ , producing the ranked paths returned to the operator.

3.2 Multi-event KIS Retrieval

A KIS request is represented as an ordered list of events. Each event may contain text, image examples, or both. For text events, SHI computes dense visual, FrameContext, and ASR scores and lexical scores from the four BM25 fields. Image examples add a visual score component for their associated event. The components are calibrated separately and combined only where the corresponding evidence is available. The result for a video is a matrix with one row per event and one column per canonical frame.

The temporal decoder searches this matrix for paths that assign one frame to each event in chronological order. Its objective combines the event scores with a configurable penalty for large temporal gaps. SHI retains multiple temporally separated paths for a video and ranks paths across the collection. The KIS result grid uses one representative frame for each returned path while preserving all aligned frame IDs, frame indexes, and timestamps for inspection.

For each executed search, SHI stores the score matrices of the returned videos, the event order, the decoded paths, and the decoder configuration in an *Evidence Snapshot*. EventTrail can therefore apply an anchor or exclusion constraint and decode the selected video again from the same scores. Query and trail revisions prevent corrections from being applied to stale search state. This feedback path changes temporal constraints without issuing another corpus-retrieval request.

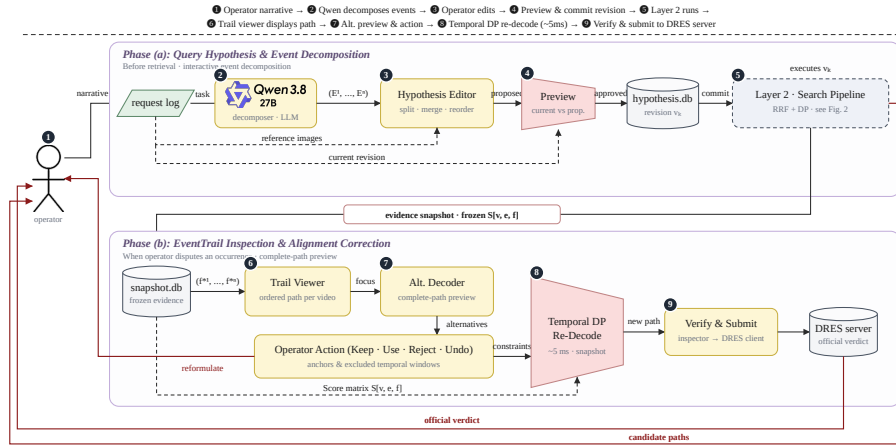


Fig. 3. The two hypothesis-level interactions in SHI. **(a)** Query Hypothesis exposes the inferred event decomposition before retrieval and supports direct structural correction. **(b)** EventTrail exposes the decoded temporal hypothesis of a retrieved video and previews complete-path alternatives before a correction is committed.

3.3 AVS and Competition Integration

AVS uses a shorter path because it does not require an ordered event sequence. A text query searches the visual, FrameContext, and ASR indexes, and SHI merges their ranked lists with reciprocal-rank fusion. A deterministic coverage stage removes nearby results from the same video and interleaves the remaining candidates by video before displaying them. Operators can collect results from successive searches in a task-scoped basket and submit the selected temporal answers as one batch.

The backend maintains the DRES session and resolves the active task before a submission. It accepts one temporal answer for KIS and one or more temporal answers for AVS, rejects answers whose type or task scope does not match the active task, and reports whether DRES recorded the submission. This validation keeps search-state identifiers separate from the canonical video and temporal coordinates used in an answer.

4 Interactive Hypothesis Correction

SHI exposes two intermediate decisions in multi-event retrieval as explicit interaction objects. Before retrieval, the *Query Hypothesis* represents how the input narrative has been interpreted as an ordered event sequence. After retrieval, *EventTrail* exposes how those events have been aligned to occurrences within a selected video. Both representations can be inspected before the user commits a correction.

4.1 Query Hypothesis

Interactive video retrieval systems commonly support query reformulation, including suggestions derived from intermediate search results and conversational refinement with relevance feedback [11, 10, 17]. Structured interfaces have also been used to express complex and temporal queries [19, 5]. SHI differs in the object being edited: the system-generated interpretation of the original narrative is itself exposed as a persistent *Query Hypothesis* before retrieval.

For a multi-event request, SHI constructs an ordered sequence

$$H_q = (E_1, E_2, \dots, E_N),$$

where each automatically generated event is grounded to a verbatim span of the original query. The original request and its inferred event sequence are shown together, allowing the user to inspect whether the decomposition preserves the intended semantic and temporal structure.

Figure 3(a) shows the Query Hypothesis editor. Users can edit an event, split it, merge adjacent events, change event order, or add a new event. Image examples can additionally be associated with individual events. In contrast to reformulating a single text query and observing the resulting ranking, these operations directly modify the event structure that will be used by the downstream retrieval and temporal-alignment stages.

Edits are previewed before they affect the active hypothesis. The interface shows the current and proposed event sequences side by side, after which the user may cancel or apply the modification. Applying a change creates a new query revision and marks results from the previous revision as outdated; a new search is started explicitly by the user. Previous revisions can also be restored through undo. This separates inspection and correction of the query interpretation from search execution.

4.2 EventTrail

Prior systems support temporal query formulation and retrieval over sequences or temporally related search conditions [6, 5, 9]. EventTrail instead exposes the *decoded temporal hypothesis* of an individual retrieved video: the current assignment of each query event to a specific occurrence in that video.

Figure 3(b) presents this hypothesis as an ordered event path, with each event associated with its selected frame and timestamp. The user can focus on any event and inspect temporally distinct alternative occurrences under the current constraints.

A central distinction of EventTrail is that an alternative is not presented as an isolated frame. For each candidate occurrence of the focused event, SHI decodes a complete chronologically valid path for the entire event sequence. The user can therefore preview the consequence of selecting an alternative before changing the active hypothesis. If other event assignments must move to preserve chronological consistency, those changes are shown together with the previewed path.

Previewing is non-mutating. The user can *Keep* the current occurrence, *Use* a previewed alternative, or *Reject* an incorrect temporal mode. Keeping or using an occurrence introduces an anchor for that event, whereas rejection excludes the corresponding

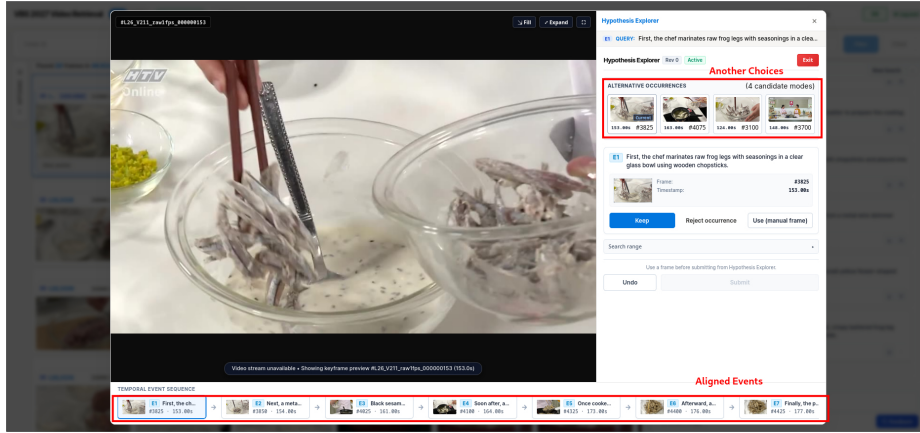


Fig. 4. The EventTrail (Hypothesis Explorer) interface. The bottom timeline presents the decoded *Temporal Event Sequence* ($E_1 \rightarrow \dots \rightarrow E_7$) for a selected video. The right panel allows the user to focus on an event (here E_1) and inspect alternative candidate occurrences (candidate modes), with controls to *Keep*, *Reject*, or *Use* an alternative frame to trigger instant path re-decoding.

temporal region. The complete path is then re-decoded under the updated constraints using the multimodal evidence retained from the original search.

Because temporal alignment is solved jointly for the complete event sequence, a local correction may alter assignments of other events. EventTrail makes these induced changes visible rather than treating each event match independently. The interaction can be repeated across events, and previous corrections can be undone when necessary.

5 Future Work and Conclusion

SHI combines multimodal retrieval, temporal alignment, and task-aware submission in an interactive system for VBS 2027. For multi-event KIS, the *Query Hypothesis* exposes the event sequence used for retrieval, while *EventTrail* exposes the aligned path within a candidate video. Operators can therefore revise the query structure or constrain an incorrect temporal match from the search interface.

Future work will evaluate these interactions under competition conditions. We plan to compare the ranked-grid baseline, Query Hypothesis, and the complete SHI workflow using task success, search time, interaction count, and re-decoding latency. Separate ablations will test event decomposition, multimodal fusion, and anchor and exclusion feedback. These measurements will show when the additional controls assist search and when their interaction cost outweighs their benefit.

Acknowledgments. The authors thank the Video Browser Showdown organizers for providing the evaluation framework and live competition environment.

References

1. Amato, G., Bolettieri, P., Carrara, F., Falchi, F., Gennaro, C., Messina, N., Vadicamo, L., Vairo, C.: VISIONE at video browser showdown 2022. In: *MultiMedia Modeling. Lecture Notes in Computer Science*, vol. 13142, pp. 543–548. Springer (2022). https://doi.org/10.1007/978-3-030-98355-0_52
2. Amato, G., Bolettieri, P., Carrara, F., Falchi, F., Gennaro, C., Messina, N., Vadicamo, L., Vairo, C.: VISIONE at video browser showdown 2023. In: *MultiMedia Modeling. Lecture Notes in Computer Science*, vol. 13833, pp. 615–621. Springer (2023). https://doi.org/10.1007/978-3-031-27077-2_48
3. Amato, G., Bolettieri, P., Carrara, F., Falchi, F., Gennaro, C., Messina, N., Vadicamo, L., Vairo, C.: VISIONE 5.0: Enhanced user interface and AI models for VBS2024. In: *MultiMedia Modeling. Lecture Notes in Computer Science*, vol. 14557, pp. 332–339. Springer (2024). https://doi.org/10.1007/978-3-031-53302-0_29
4. Gasser, R., Arnold, R., Faber, F., Schuldt, H., Waltenspil, R., Rossetto, L.: A new retrieval engine for vitivr. In: *MultiMedia Modeling. Lecture Notes in Computer Science*, vol. 14557, pp. 324–331. Springer (2024). https://doi.org/10.1007/978-3-031-53302-0_28
5. Heller, S., Arnold, R., Gasser, R., Gsteiger, V., Parian-Scherb, M., Rossetto, L., Sauter, L., Spiess, F., Schuldt, H.: Multi-modal interactive video retrieval with temporal queries. In: *MultiMedia Modeling. Lecture Notes in Computer Science*, vol. 13142, pp. 493–498. Springer (2022). https://doi.org/10.1007/978-3-030-98355-0_44
6. Heller, S., Gasser, R., Illi, C., Pasquinelli, M., Sauter, L., Spiess, F., Schuldt, H.: Towards explainable interactive multi-modal video retrieval with vitivr. In: *MultiMedia Modeling. Lecture Notes in Computer Science*, vol. 12573, pp. 435–440. Springer (2021). https://doi.org/10.1007/978-3-030-67835-7_41
7. Hezel, N., Schall, K., Jung, K., Barthel, K.U.: Efficient search and browsing of large-scale video collections with Vibro. In: *MultiMedia Modeling. Lecture Notes in Computer Science*, vol. 13142, pp. 487–492. Springer (2022). https://doi.org/10.1007/978-3-030-98355-0_43
8. Jäckl, B., Khan, O.S., Verner, B.V., Vopalkova, Z., Schlegel, U., Keim, D.A., Lokoč, J.: What drove success at the 15th video browser showdown? a comprehensive interaction-logging analysis. In: *Proceedings of the 2026 International Conference on Multimedia Retrieval*. pp. 1730–1739. ACM (2026). <https://doi.org/10.1145/3805622.3810635>
9. Khan, O.S., Sharma, U., Marcelino, G., Rudinac, S., Jónsson, B.P.: Exquisitor at the video browser showdown 2026: Temporal queries revisited. In: *MultiMedia Modeling. Lecture Notes in Computer Science*, vol. 16415, pp. 245–251. Springer (2026). https://doi.org/10.1007/978-981-95-6963-2_27
10. Khan, O.S., Zhu, H., Sharma, U., Kanoulas, E., Rudinac, S., Jónsson, B.P.: Exquisitor at the video browser showdown 2024: Relevance feedback meets conversational search. In: *MultiMedia Modeling. Lecture Notes in Computer Science*, vol. 14557, pp. 347–355. Springer (2024). https://doi.org/10.1007/978-3-031-53302-0_31
11. Lokoč, J., Vopálková, Z., Dokoupil, P., Peška, L.: Video search with CLIP and interactive text query reformulation. In: *MultiMedia Modeling. Lecture Notes in Computer Science*, vol. 13833, pp. 628–633. Springer (2023). https://doi.org/10.1007/978-3-031-27077-2_50
12. Lokoč, J., Vopálková, Z., Stroh, M., Buchmüller, R., Schlegel, U.: PraK tool: An interactive search tool based on video data services. In: *MultiMedia Modeling. Lecture Notes in Computer Science*, vol. 14557, pp. 340–346. Springer (2024). https://doi.org/10.1007/978-3-031-53302-0_30
13. Ma, Z., Ngo, C.W.: Interactive video corpus moment retrieval using reinforcement learning. In: *Proceedings of the 30th ACM International Conference on Multimedia*. pp. 296–306. ACM (2022). <https://doi.org/10.1145/3503161.3548277>

14. Rossetto, L., Schuldt, H., Awad, G., Butt, A.A.: V3C: A research video collection. In: MultiMedia Modeling. Lecture Notes in Computer Science, vol. 11295, pp. 349–360. Springer (2019). https://doi.org/10.1007/978-3-030-05710-7_29
15. Schall, K., Bailer, W., Barthel, K.U., Carrara, F., Lokoč, J., Peška, L., Schoeffmann, K., Vadicamo, L., Vairo, C.: Interactive multimodal video search: An extended post-evaluation for the VBS 2022 competition. *International Journal of Multimedia Information Retrieval* **13**(2), 15 (2024). <https://doi.org/10.1007/s13735-024-00325-9>
16. Schoeffmann, K., Nasirihaghighi, S.: DiveXplore at the video browser showdown 2024. In: MultiMedia Modeling. Lecture Notes in Computer Science, vol. 14557, pp. 372–379. Springer (2024). https://doi.org/10.1007/978-3-031-53302-0_34
17. Sharma, U., Khan, O.S., Rudinac, S., Jónsson, B.Þ.: Exquisitor at the video browser showdown 2025: Unifying conversational search and user relevance feedback. In: MultiMedia Modeling. Lecture Notes in Computer Science, vol. 15524, pp. 264–271. Springer (2025). https://doi.org/10.1007/978-981-96-2074-6_31
18. Video Browser Showdown Organizers: Call for papers (VBS 2027). <https://videobrowsershowdown.org/call-for-papers/>, accessed 21 September 2026
19. Wu, J., Nguyen, P.A., Ma, Z., Ngo, C.W.: Sql-like interpretable interactive video search. In: MultiMedia Modeling. p. ??? (2021). https://doi.org/10.1007/978-3-030-67835-7_34